

ParkIt: Smart Parking Management Platform for Vehicle Entry, Allocation, Occupancy Monitoring, and Exit Management

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Abstract—Efficient parking management has been a persistent concern for communities, educational institutions, commercial establishments and urban infrastructure due to increasing vehicular population and limitations of traditional parking systems. Existing parking management solutions cater to different aspects of parking management leading to disjointed workflows and sub-optimal utilization of available resources. In this paper, we describe ParkIt, a unified parking management platform that aims to digitalize and automate the entire parking lifecycle for a parking ecosystem consisting of vehicle owners, visitors, security officers and parking administrators. ParkIt combines a mobile application integrated with a centralized management console to enable parking oversight and control. The platform prototype provides a blueprint for digitalizing parking systems using a distributed-registrar, allowing for efficient parking management at a large scale, spanning urban infrastructure, commercial, campus and residential setups, among others. Our experimental results confirm the feasibility of adopting integrated parking management platforms towards efficient utilization of parking spaces.

Keywords— *Smart Parking, Parking Management Platform, Vehicle Management, Occupancy Monitoring, Real-Time Parking Operations.*

I. INTRODUCTION

Introduction Rapid Vehicle Population Growth Increasing vehicle ownership rates across residential communities, educational institutions, commercial complex, healthcare facilities and public infrastructure has resulted in a spurt in demand for effective parking management. While modern transportation systems are making progress, the majority of parking management systems continue to use manual supervision, paper-based reservation systems, or integrated digital systems that handle isolated features of parking management. These shortcomings often result in inefficient utilisation of space, vehicle movement, administrative

inefficiencies, and substandard user experience. Existing parking management solutions Several parking solutions have been proposed in recent years to simplify parking management and improve parking access. In these existing solutions, each parking management feature is either implemented independently or as a functionality within a larger integrated system that is independent of other parking management features. As a result, parking users, security personnel, and administrators depend on different parking systems to fulfil parking requirements, resulting in communication gaps and administrative inefficiencies. Requirements for a parking management system, the requirement of a parking management system is to address issues in the current systems, and manage the entire lifecycle of a parking operation, from the registration of vehicles and granting entrance access, to allocations, managing occupancy and administrative oversight, and exit management. It would require providing visibility of parking resources between the users and administrators in real-time while reducing the dependence of the users, security personnel, and administrators on manual interaction by ensuring transparency in the entire process. Overview of solution Bringing together of several parking services into one scalable platform: A parking management platform that brings together several parking features into one scalable platform that would digitalize the entire parking lifecycle - from registration of vehicles/visitors to managing operational costs, controlling parking spaces, analysing parking occupancy, managing administration, and tracking revenue collection in a single platform would improve efficient usage of parking spaces and improve parking management. This paper presents the design and implementation of a parking management platform called ParkIt that combines parking management features along with a modular sensor and camera infrastructure to enable parking management with a single control platform. ParkIt offers a comprehensive solution for managing the entire parking process with real-time surveillance, automated parking allocation, and centralized control. The application ensures better parking efficiency, promotes convenience for users and supports the growth of the advanced parking system.

II. RELATED WORK

Parking management has attracted much attention due to the limitations in capacity and the pressing need to maximize parking resource usage. [1], [5] Existing smart parking systems utilize mobile apps, cloud servers, internet of things (IoT) devices, occupancy sensors, etc., to facilitate parking access and make operations more efficient. [1], [3], [5] These existing smart parking systems are offering benefits of parking reservation, parking slot, and occupancy monitoring, and administration monitoring, but they are mostly standalone applications without fully integrated parking operational perspectives at the interface of parking users, parking administration, and parking infrastructure. [5], [7]. Consequently, parking system's operational process is fragmented and lacks transparency, highlighting the necessity of providing a single, integrated parking platform. ParkIt is such a platform that can operate the entire life cycle of parking including vehicle management, parking allocation, occupancy detection, administration, and parking sessions management.

A. Smart Parking Management Systems

Smart parking management system research has moved from an emphasis on hardware-based, on-street parking infrastructure towards integrated, centralized parking management system software platforms. [1], [5]. Conventional, microcontroller-based on-street parking systems rely on the networking of individual ultrasonic, infrared and magnetic loop occupancy sensors, which only monitor the presence of a vehicle to ascertain available counts. [1], [3], [8] Without a central management structure, such systems lack cross-platform front-end support for end-users as well as remote backend management for administration. [1], [5] Current solutions are focused on centralized management that enables system administrators to manage and monitor parking status, user activity, and vehicle data through an application interface, but requires highly coordinated, inter-related services for end-users, management and the infrastructure. [2], [5], [6].

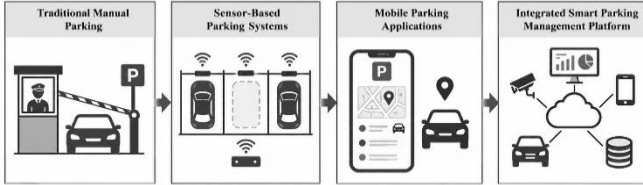


Fig 1. Evolution of Parking Management System

As shown in Fig. 1, one can perceive the development of parking management systems follows a linear transition through four paradigms. It goes from the conventional manual parking operation with staff and paper/ticket-based operation, to sensor-based parking systems with localized hardware monitoring, to driver-facing mobile parking apps with remote visuals, then finally to interconnected smart parking management platform architecture in the form of ParkIt with integrated hardware inputs, driver-facing interfaces and administrator controls.

B. Mobile-Based Parking Applications

The proliferation of mobile phone use has facilitated the development of driver focused parking apps. [4], [9]. Early mobile parking apps were primarily directory applications that could report static location and operating information. [9], [10]. While much has been achieved in the development of driver facing mobile applications, many such applications are still not integrated with facility side gate control systems and management software. [4], [7]. One of the most

challenging issues with a number of existing mobile applications is an information gap created by the mismatch between mobile reservation status and actual real-world availability, resulting in over- or under-occupancy of parking slots. [5], [10]

C. Occupancy Monitoring and Parking Information Systems

An occupancy monitoring solution enables real-time visualization of parking bay utilization, and informed parking use decisions for users and operators alike. In a real-world parking environment, sensing itself is only one part of the solution. The detection data must be interpreted, stored and shared by an admin infrastructure to enable many of the uses applications sought. [1], [5], [6]. Few admin dashboards feature a visual map of space utilization, requiring visual operators to scry on text logs or database tables. Visitor management is often absent from digital tools, resulting in security personnel recording on paper, in addition to using digital gate technology, introducing possible audit trails and security concerns. [7]

D. Research Gap and Motivation

A literature review of research in the parking systems domain showcases a significant disconnect between detection technologies on one hand and management frameworks and software on the other. Much of the research is focused at isolated sub-systems (sensor calibration, detection models and user interface design) and much less on system level integration needed to connect these components and make a working whole. The relatively disconnected parking system components, in terms of mobile applications and direct vehicle access, real-time data pipeline, dynamic slot assignment, visitor permission and digital operator dashboards, can be challenging to connect into a single, unified system. [4], [5], [7]. Hence, ParkIt is envisioned as a scalable, yet integrated parking management system framework that brings together the entire parking ecosystem - from vehicle registration, entry permission, slot management, occupancy management, visitor management, admin supervision and vehicle exit into a single service.

Feature	Traditional Systems	Mobile Parking Apps	ParkIt (Our Application)
Vehicle Management	Manual	Limited	Integrated
Parking Allocation	Manual	User-Based	Automated
Occupancy Monitoring	Limited	Not Available	Real-Time
Administrative Control	Minimal	Basic	Centralized Dashboard
Visitor Management	Manual Records	Not Supported	Digital Workflow
Parking Lifecycle Support	Partial	Partial	End-to-End

Table I. Comparison of Existing Parking Solutions and ParkIt

According to Table I, a side-by-side comparison of core system capabilities shows the unique functional space occupied by the ParkIt platform compared to current architectural modes: Sensor-based local systems provide local counting and display, but have no tools for system admins or drivers. The ParkIt platform is a single integrated product for end-to-end lifecycle management including real-time data sync, mobile and centralized dashboards, the digital visitor workflow, slot allocation, and centralized data management.

III. SYSTEM DESIGN OVERVIEW

The ParkIt system was developed as an integrated software platform to manage the full parking lifecycle across enterprise and campus operations. Instead of systemically isolating vehicle access, space distribution, occupancy measurement and clear-out functions, the ParkIt platform combines these operational elements into a cohesive, centralized architecture, connecting end-user mobile apps, administrative controls, and back-end systems in a shared operational loop that provides constant, real-time management feedback for drivers and operators.

A. Overall Platform Architecture

The system architecture of ParkIt consists of five partitions extending across the relational database, business logic, and client applications. Client applications, business logic partitions, and database are all decoupled through the use of separate API routes, transactional API command windows, duplex event streams, and stateless data persistence. The client partition is further divided into an end user and facility operator interface built on Flutter and React.js respectively, while the communication channels between clients, server, and data store use HTTP/2, web sockets, and the Node.js or Express.js single process server gateway to support requests routing and API event relays. Logical units of core parking transaction processing are modularized in a set of functional server services for vehicle and visitor management, entry management, automated slot assignment, payment processing, and exit management, while the core data store across the entire platform is managed in a relational Postgres database coupled to the Prisma ORM data access API.

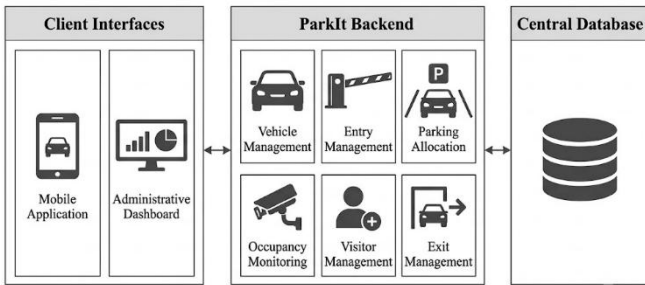


Fig 2. Evolution of Parking Management System

B. Stakeholders and User Roles

ParkIt is designed to meet the needs of all the various stakeholders who are involved in ordinary parking operations. The system mainly caters to registered drivers, visitors, and facility administrators, each of whom uses the system via dedicated interfaces and features that are specific to their role. Since all these interactions are brought together in one ecosystem, the platform enables more efficient coordination, better visibility, and smoother parking management. Drivers who are registered can use the mobile application to look after vehicle details, get access to parking services, receive assignments for parking spaces, and keep an eye on their current parking sessions. The application features a convenient and user-friendly interface which allows drivers to access parking-related information in real time and at the same time decreases the need for manual interactions when carrying out the entry and exit processes. People are dealt with via a controlled onboarding procedure which is intended to give them temporary access to parking. Whether or not pre-authorization is required by the facility, visitors can either be pre-authorized by a host or sign in at the specified entry points before their parking rights are granted. This process aids in keeping visitor records accurate while at the same time ensuring secure and efficient access control in the parking

facility. Facility administrators and security staff access the system via a central web-based dashboard which acts as the system's operational control centre. This dashboard gives a real-time view of parking occupancy, the movement of vehicles, the activities of visitors, and the records of parking sessions. Through a single management interface, administrators are able to keep an eye on facility operations, deal with incidents relating to parking, handle access requests, and monitor the general level of parking usage. Since it meets the needs of all the stakeholders, ParkIt allows for coordinated parking operations and improves the efficiency of facility management.

C. System Workflow

ParkIt's operational workflow is based on a structured parking lifecycle which handles the movement of vehicles from registration and obtaining entry authorization through to parking space allocation, the monitoring of occupancy, and the processing of exits. The process starts when a user registers via the mobile application and links one or more vehicles to the account. When the user reaches the parking facility, the vehicle goes through an authentication and verification procedure to ensure that it is eligible for access. After having been verified, the system looks at the current level of parking availability and then assigns an appropriate parking space according to the predefined allocation rules and the current occupancy status.

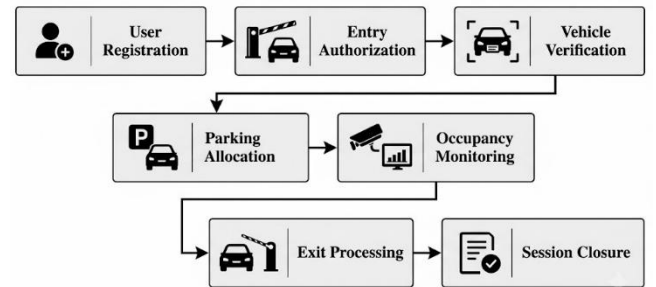


Fig 3. ParkIt System Workflow

Once a parking space has been assigned, an active parking session is set up and kept going for the entire duration of the vehicle's time in the facility. The occupancy updates and session details are constantly synchronized with the central backend, so that both users and administrators get accurate real-time parking information. The mobile application gives users access to their assigned parking space and to the session details, while the administrative dashboard shows the latest occupancy conditions throughout the facility. On leaving the parking facility, the exit management module obtains the current parking session, logs the departure event, and ends the session lifecycle. The parking space in question is then made available and shown as being free for future use. This end-to-end process allows for smooth coordination between users, administrators, and the parking infrastructure, thus ensuring efficient parking operations, better use of space, and consistent visibility throughout the platform. Moreover, the workflow facilitates the flow of data between all components of the system constantly, ensuring that parking records are correct and updated throughout all the lifecycle of parking. The workflow connects user interaction, parking operations, and system management in a single platform ecosystem, improving the effectiveness of operations and making parking easily accessible to everyone involved. Also, the modular design of the workflow ensures the adaptability of the system to different conditions of parking in residential, educational, commercial, and large public spaces. The workflow also establishes a scalable base for any next-level improvements, advanced parking services and predictive analytics are smoothly integrated on future.

IV. SYSTEM IMPLEMENTATION

The ParkIt platform is built as a comprehensive parking management system to control and manage vehicle entry, parking allocation, parking sessions and exit. It has two main interfaces: an administrative dashboard for the facilities' management and a mobile application for the customers' use. The two interfaces are linked and provide consistent information exchange including data on vehicle entry, parking spots, parking sessions and exits.

A. Administration Dashboard

The Administrative Dashboard is the main hub of the ParkIt system, designed to allow administrators to view all of the parking in the ParkIt system from one user-friendly, web-enabled portal. The Administrative Dashboard is designed to allow for quick management of parking operations by providing information related to parking spot occupancy, current parking sessions, active vehicles, live video streaming, parking capacities, and other system status updates on the web-based portal. The dashboard presents real-time reports, metrics, and parking system insights in a clear, structured manner giving administrators a quick insight to the parking lot health. Real-time communication modules are integrated with the dashboard to inform any update in occupancy, vehicle movements, and system events instantly. In addition to tracking parking usage, the Administrative Dashboard allows parks an all-in-one dashboard and menu to manage parking infrastructure and workloads. Providing one central place to view real-time parking information, administrator tasks, configuration, and reporting information, the Administrative Dashboard is a single and intuitive point of access for parks and optimized the overall efficiency of the system and thorough visibility into the entire ParkIt ecosystem.

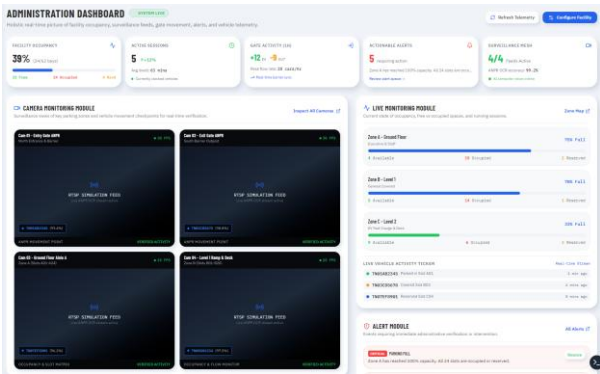


Fig 4. Admin Dashboard

B. Mobile Application

The ParkIt mobile app is the main user interface of the smart parking platform being proposed, allowing users to use the parking services and look after all parking-related matters directly from their mobile devices. It has been developed with the help of Flutter and offers a single, user-friendly environment which makes the parking process easier while at the same time ensuring smooth communication with the backend services. The home screen functions as the main dashboard of the application, giving users a summary of the parking services available and enabling them to quickly reach the key features. Users can request parking services and see the parking options that are available prior to starting a parking session. The system, taking into account the parking availability and the allocation rule it has established, assigns an appropriate parking space to the user. After the allocation has been carried out, the booking details are sent to the backend and an active parking session is set up. This automated method of allocating parking spaces cuts down the

time needed to search for parking and helps to improve the way parking is used in the facility. In order to enable secure verification of vehicles and control of access, the application includes a system based on QR codes. The QR code produced acts as a digital identifier and is used when people enter or leave. The Screen for Session Details gives users full information about their current parking session, showing details such as the parking space assigned, the status of the session, and other parking-related information via this interface. Using the Profile module, users can look after their personal information and the details of the vehicle they have registered in the application. The mobile app improves convenience for users by combining in one place the functions of booking, parking access, session monitoring, and profile management.



Fig 5. Home Screen

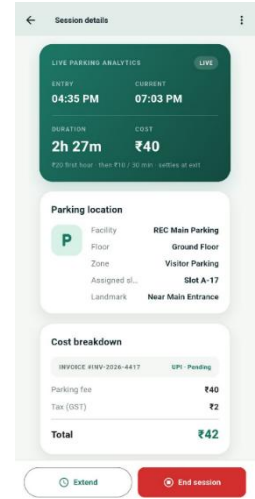


Fig 6. Session Details

C. Parking Allocation and Occupancy Management

The parking allocation component makes the connection between the parking request made by the user and the availability of spaces in the facility. Once a successful request is made or the user gains access to the facility, the system assigns a parking spot to the respective vehicle. This allocation is made evident in the user's application, such that the user can be able to tell the parking spot allocated to him/her, while at the same time, there is an update on the dashboard as far as the availability of spots is concerned. With the vehicles coming in and out of the parking spots, the status of the vehicles is updated accordingly.

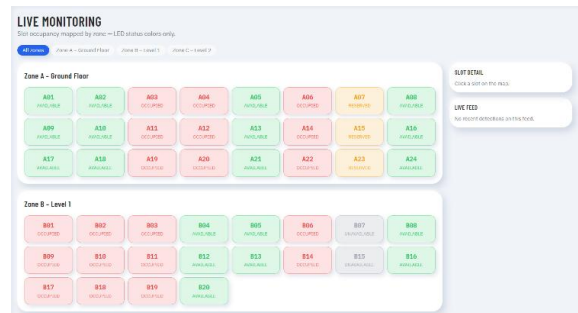


Fig 7. Live Monitoring

D. Parking Session Management

The parking session can be defined as the time between when the parking spot has been allocated and when the vehicle has successfully parked. ParkIt holds all the details about the session that is active so that the status of the session is visible to both the user and the administrator. The details of the parking session for the user are visible via the mobile application, while the active parking session is visible to the

administrator via the dashboard. The status of the session keeps changing with the progress of the vehicle in the parking cycle.

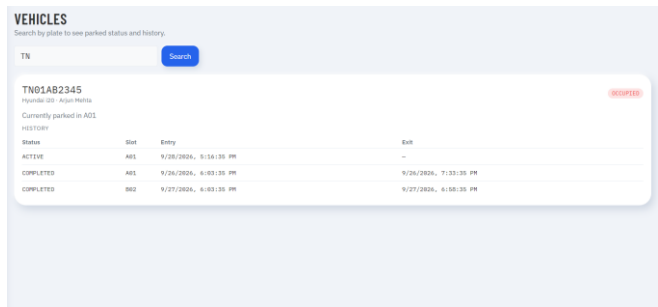


Fig 8. Vehicle Management

E. Entry and Exit Management

These two functions take care of the movements of vehicles on entry and exit from the parking lot. In case of the entry process, the system checks the required vehicle and parking data prior to approving the parking activity. After allocating a parking slot, the vehicle is now involved in an ongoing parking activity with regard to its occupancy status. The exit function sees the completion of the parking activity, release of the occupied slot, and the change in the occupancy status shown on the platform. In essence, this helps track the digital occupancy status based on the movement of the vehicle, and there will be no need for constant manual updating of parking availability status.

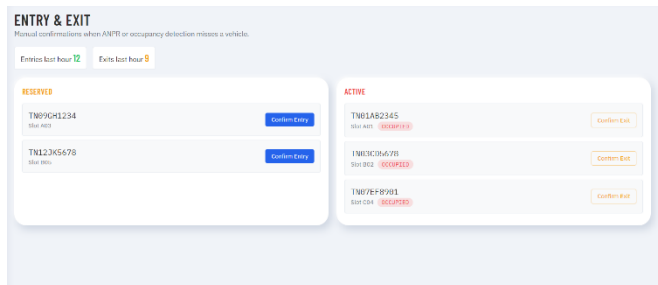


Fig 9. Entry and Exit Management

F. Integrated Platform Operation

ParkIt platform that is already integrated unites these two roles and enables them to cover the entire parking cycle. Users are able to use the mobile app, ask for parking, get an available space, track their active parking session, whereas administrators are capable of seeing at the same time such things as occupancy, cameras, cars moving, entry and exit processes, alerts and reports using the dashboard. All the changes made during parking process will be shown on the corresponding interface, and it will enable users as well as facility owners to deal with the existing situation of parking.

V. RESULTS AND DISCUSSION

In addition to that, the testing procedure examined the integration of different elements of the system including backend services, admin dashboard, mobile app, and computer vision technology in order to achieve uninterrupted flow of data. Various real-life scenarios were tested for real-time parking allocation, space occupancy updates, vehicle tracking, and session syncing. The results showed that all tested modules operated as intended by giving accurate parking information at all stages of the parking process. The system proved to operate flawlessly with respect to the continuous operation which means that it supported all parking processes functionally as expected. The results obtained reflected consistent quality performance with

minimal human effort used while executing the parking processes. All crucial functions were performed as expected and gave consistent results throughout testing process. Passing the tests means that ParkIt platform is fully operational and reliable.

A. Functional Validation

The ParkIt system was tested in the parking areas of Rajalakshmi Engineering College, the site consisting of four sections allocated for various categories of users, including visitors, teachers, students, and administrative staff. Validation testing was carried out by carrying out the main business processes via the mobile application and the administration panel. It was found that the procedures involving vehicle registration and entrance management, parking allocation and the monitoring of occupancy, visitor handling, the operation of the mobile application, the administrative functions, and exit management were all working correctly. In order to check that all the platform modules carried out their functions as expected and stayed in sync, a number of test cases were performed. The results show that the platform is able to effectively support everyday parking operations while at the same time providing administrators with operational visibility. The main functional modules have been validated as set out in Table II.

Module / Workflow	Validation Objective	Result
Vehicle Registration	Verify user and vehicle onboarding process	Successfully Validated
Entry Management	Validate vehicle authentication and access control	Successfully Validated
Parking Allocation	Verify automatic assignment of available parking spaces	Successfully Validated
Occupancy Monitoring	Validate parking occupancy status updates	Successfully Validated
Mobile Application	Validate user interaction and parking service accessibility	Successfully Validated
Administrative Dashboard	Verify monitoring, management, and reporting functions	Successfully Validated
Exit Management	Validate session closure and parking space release process	Successfully Validated

Table II. Functional Validation Results of the ParkIt

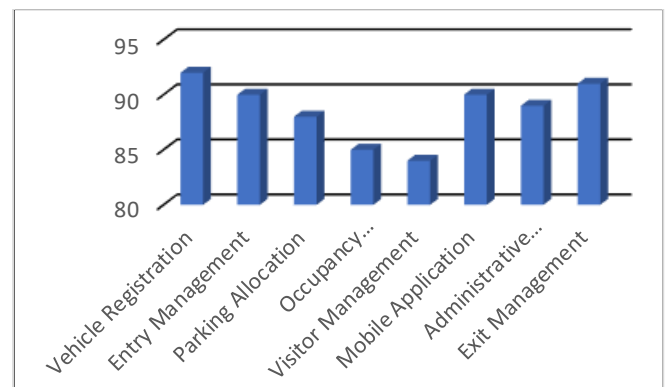


Fig 10. Functional Testing Outcomes of ParkIt

B. Workflow Validation and Operational Observations

Testing covered the entire process starting from user registration to the time the vehicle left the facility. In the testing process, users were able to register their vehicles using the mobile application and continue with the parking process. They were also given the available parking spaces from the

platform, and they would choose an allocation that would be registered into their parking process. This helped reduce the effort of searching for parking places by the users. Another thing tested in the process was the visitor handling. This could be done through the administrative interface where the authorized personnel could register visitors' vehicles and give them access details using the platform. This was an effective way of managing visitors through the digitalization process because it allowed the tracking of visitor activities within the parking process. Lastly, testing of administrative controls could be done through changing the operational states of the selected parking spaces. This could be done by the administrators where they would change the state of a parking space to unavailable when it is not suitable for allocation due to some reasons such as blocking or under maintenance. The system maintained consistent parking status across both user and admin interfaces.

C. System Integration and Real-Time Synchronization

The combination of the mobile application and administrative platform were assessed while monitoring for changes to parking conditions during concurrent parking activities. Changes to parking-space status and active parking sessions, as well as administrative status, appeared across the relevant portions of both platforms, giving users and admins a consistent view of parking conditions. We also assessed visibility into changes due to parking allocation, completion of parking sessions, and admin parked space release. During parking sessions that began concurrently, the implemented platform accurately tracked changes to individual parking space status and did not issue the same available parking space to multiple concurrent parking sessions. The updated parking spaces appeared on the interface without requiring the user to refresh the page for additional parking sessions. These results demonstrate that the platforms made with the ParkIt infrastructure can successfully coordinate parking sessions and admin operations while providing a clear view of parking status.

D. Discussion

The assessment shows that the ParkIt system successfully bridges between the user interaction with parking to control at the parking facility level. The mobile application is the driving interaction point with the users, while the admin dashboard manages the monitoring and configuration, managing the visitors and their vehicles, raising alerts, and the reporting. The functional assessment and workflow watching showed that these two components could be used together through all major phases of a parking lifecycle. The results also emphasised the usefulness of having a shared, continuously updated parking state in mind, especially when parking information is updated based on vehicles and staff move around the parking. Additional assessments at larger parking facilities over a longer period with more real-life traffic will help the assessment. The centralized management approach improved operational visibility and simplified parking administration tasks. Overall, the evaluation results indicate that ParkIt is capable of supporting efficient, scalable, and reliable parking management operations. These findings validate the practicality of the proposed solution for real-world smart parking applications.

VI. FUTURE SCOPE

The ParkIt platform offers a scalable base for the next generation of smart city and facilities management applications. Development will, in future, expand in the following areas. Planned field areas include, EV Charging Integration. The parking allocation engine can be enhanced to

take into account the battery state of charge, plug type compatibility, and charger supply availability at the time of allocating parking spots. A predictive analytics layer can be built that uses historic parking data to predict vehicle arrivals, occupancy peaks, and zone level demand. In commercial deployments, a dynamic pricing module and a secured payment gateway can be used to support time-based parking tariffs based on facility demand. Additional enhancements can involve auto-calibration of cameras, which is achieved using such tools as diagnostic functionality for camera misalignment, nonoptimal lighting or other video feed problems. Vehicle-to-Infrastructure (V2I) Communication can also be investigated for connected vehicles to provide parking and wayfinding data directly to the ParkIt backend system. For the future, the platform can be expanded to support auto-pilot parking by providing optimal parking spot coordinates and wayfinding data to autonomous vehicles.

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